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Food management in tourism: Reducing tourism's carbon 'foodprint'

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A B S T R A C T

Food production and consumption have a range of sustainability implications, including their contribution to global emissions of greenhouse gases (GHGs). As some foodstuffs entail higher GHG emissions than others, managing their use in tourism-related contexts could make a significant contribution to climate change mitigation. This article reviews the carbon intensity of selected foods and discusses how foodservice providers could adapt their practices. It shows that even though food management could substantially reduce the GHG emissions of foodservice providers, its application is currently hampered by the complexity of food production chains and a lack of dependable data on the GHG intensity of foodstuffs. Nevertheless, it is possible to make a number of recommendations in respect of how foodservice providers can better purchase, prepare and present foods. Further research is now needed to refine and extend our understanding of the contribution that food management can make to reducing tourism's carbon 'foodprint'.

1. Introduction

Food production has a wide range of sustainability implications, including land conversion and the associated loss of biodiversity and ecosystems (Lawton & May, 1995; Pimm, Russell, Gittleman, & Brooks, 1995; Vitousek, Mooney, Lubchenco, & Melillo, 1997); changes in global biogeochemical processes, such as nitrogen cycles (Vitousek, Aber, et al. 1997); water consumption (Chapagain & Hoekstra, 2007, 2008; Hoekstra & Chapagain 2007); the use of substances potentially harmful to human health, such as pesticides, herbicides and fungicides (Bhalli et al., 2009; Koutros et al., 2008); ethical questions, such as those relating to animal welfare and genetically modified organisms (Zollitsch, Winkler, Waiblinger, & Haslberger, 2007), and the foodservice sector's contribution to the global emissions of greenhouse gases (GHGs) relating to agriculture, food processing, transport and the preparation of meals.

Food consumption, meanwhile, is widely recognised to be an essential part of the tourism experience (Boniface, 2003; Hall & Sharples, 2008; Hall, Sharples, Mitchell, Macionis, & Cambourne, 2003; Hjalanger & Richards, 2002). Locally distinctive food can be important both as a tourism attraction in itself and in helping to shape the image of a destination (Cohen & Avieli, 2004; Hall et al. 2003; du Rand & Heath, 2006). A number of recent articles have emphasised the potential for local food experiences to contribute to sustainable development, help maintain regional identities and support agricultural diversification (Clark & Chabrel, 2007; Everett & Aitchison, 2008; Knowd, 2006; Sims, 2009). Other writers have identified the market potential for 'gastronomic tourism', arguing that it has strong sustainability credentials (Hall, 2003; Hall et al. 2003; Hjalanger & Richards, 2002; du Rand & Heath, 2006). These findings suggest that both the production and consumption of food have an important role to play in promoting sustainable tourism development.

As writers such as Telfer and Wall (2000) and Hunter (2002) acknowledge, however, the sustainability implications of food consumption and production have so far received only scant attention in the tourism literature. Most of the existing research has concentrated on tropical islands, such as Jamaica, Zanzibar and Indonesia (Bélisle, 1984; Gössling 2001; Telfer & Wall, 2000). A major focus of such research has been to investigate the potential for conflict between agriculture and tourism in the competition for land, labour and capital (e.g. Bélisle, 1983, 1984). Such conflicts can, however, arise in any part of the world where tourism competes

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heavily for scarce resources, and result in a wide array of economic, socio-cultural, political and environmental problems (Gössling, 2001). For example, the appropriation of resources by tourism can necessitate the importation of food supplies (Telfer & Wall, 1996). Ultimately tourism might even bring about the demise of agriculture by out-competing it for resources (Weaver, 1988).

The climate change dimension of food has therefore not been covered in great depth in the tourism literature. No study has yet been published that considers in detail the inter-relationships implicit to tourism-related food consumption, energy use and GHG emissions. A few studies have attempted to incorporate food into the total ecological footprint for tourism at a specific destination, or in particular hotels (e.g. Peeters & Schouten, 2006; WWF-UK, 2002). However, such studies tend to be limited by the vast amount of data required to develop sophisticated food management carbon inventories. The contribution made by food consumption to tourism's carbon footprint – what we might call tourism's 'carbon foodprint' – is therefore still unknown. This is illustrated by the comments of Hunter and Shaw (2007: pp. 54–55), who suggest that:

> Eating and drinking more locally/nationally produced food and beverage than residents… is likely to have a proportionately small impact on the destination EF [ecological footprint]. Where food and drink are imported for tourists, however, and the energy embodied in these products is therefore high, tourism might generate a large additional EF at the destination.

In spite of the evident paucity of empirical studies that have examined tourism's carbon foodprint, it is clear that food production and consumption are key issues for climate change mitigation. Agriculture accounted for between 10% and 12% of total anthropogenic GHG emissions in 2005 (Smith et al., 2009). Most problematic are the sector's emissions of methane (CH_4) and nitrous oxide (N_2O), to which agriculture contributes 47% and 58% respectively of the global anthropogenic total (Smith et al., 2009). These are both potent GHGs (IPCC, 1995). Emissions from global fisheries would have to be added to those from agriculture, but no estimates of the emissions attributable to global fisheries presently exist (however, see Tyedmers, 2001). Food production and consumption also embraces packaging, retailing, transport and preparation, all of which add considerably to overall emissions from this sector. A calculation made on an end-use-consumption basis revealed that food consumption accounts for more than 20% of Norway's total GHG emissions (Hille, Sataøen, Aall, & Storm, 2008). This figure might be considered representative of a large number of industrialised countries.

It is argued that GHG emissions will have to be reduced considerably from their present levels in order to avoid climate change of a magnitude that will have serious negative consequences for society (IPCC, 2007; Stern, 2006). Emissions from food production are, however, expected to increase for two main reasons. First, a growing world population demands more food (Garnett, 2009). Secondly, changes in dietary preferences towards higher-order foods can be expected to increase GHG emissions, with trends towards more intensive production of beef, poultry and pork being observable in many parts of the developing world (Smith et al., 2009). A growing demand for meat requires the greater use of irrigation and fertiliser for fodder production, which in turn increases the demand for energy. It also induces changes in land use, for instance from forest to grassland: a process that inevitably leads to CO_2 emissions (Smith et al., 2009). Food production and its analogue, food consumption, are consequently of critical importance in the current and future development of GHG emissions. Smith et al. (2009) discuss a wide range of measures to reduce production-related emissions through agricultural management techniques, but there is also potential for the management of consumption activities to contribute to the mitigation of climate change. This article sets out to evaluate the role of tourism in food consumption, as well as to investigate the sector's potential to contribute to climate change mitigation through the use of consumption-oriented food management practices.

2. Food, tourism and climate change

Tourism is of relevance in food consumption because of the enormous amount of food that is prepared by the various foodservice providers involved in catering for tourists in both leisure and business contexts, including food stalls, restaurants, cafeterias, canteen kitchens, fast-food chains, diners, trains, aircraft, ferries and cruise ships. According to UNWTO–UNEP–WMO (2008), almost 25 billion tourist days were spent in 2005. At an average of three meals per tourist per day, this corresponds to roughly 75 billion meals per year, or just over 200 million meals per day. It can be assumed that the vast majority of these meals will be prepared by foodservice providers.

The provision of food by foodservice establishments has considerable relevance to climate change mitigation for several practical reasons. First, a relatively small number of foodservice companies influence the food consumption patterns of a comparatively large number of people. It is these companies that have the power to influence the sustainability of the food provided to tourists through their everyday business decisions. For example, the decision by Scandic hotels only to provide organic and fairly traded coffee in their hotels in Sweden has repercussions for the sustainability of millions of cups of coffee served every year (Scandic, 2009). Secondly, foodservice establishments can cook food more efficiently, as the simultaneous preparation of many meals typically entails lower energy use per meal (Carlsson-Kanyama, Pipping Ekström, & Shanahan, 2003). At the same time, foodservice establishments may also generate high amounts of food waste (Swedish Environmental Protection Agency, 2008). If the proportion of food waste is greater when an individual dines out compared to when they cook their meal at home, this will tend to increase tourism's carbon foodprint. Thirdly, foodservice providers may have considerable influence on the globalisation of the food industry. Buying food products from retailers seeking to reduce costs, for instance, increases the pressure on the global food sector to provide cheap foodstuffs. This, in turn, encourages the industrialisation of food production, which Vos (2000) argues has led to many of its current problems, including outbreaks of swine fever and bovine spongiform encephalopathy.

These are just some examples of the sustainability implications of food consumption in the context of tourism. By applying food management practices and making more informed choices about the purchasing, preparation and presentation of their food, foodservice providers could contribute to a more sustainable system of food production and consumption. In doing so, they may assist in mitigating global GHG emissions. This potential has indeed been realised by some governments, organisations and retailers seeking to support more climatically sustainable food consumption. Tesco, the British supermarket giant, has recently started to roll out a programme of carbon labelling, in which the CO_2-equivalent (CO_2-e) content of 100 of its own-brand products is indicated on their packaging (Tesco, 2009). A number of organisations, in some cases on behalf of governments, have also developed food calculators to help inform customers about the options available to them in choosing low-carbon food (one example is Food Carbon, 2009). Meanwhile the Norwegian Technology Council (2008) recently recommended that government organisations should seek to develop a better knowledge of the GHG emissions associated with food products and to adopt an internationally standardised method for calculating them. This would enable more effective 'climate labelling', which in turn would aid consumers in adopting climate-friendly consumption patterns.

3. Towards climatically sustainable food management

Any analysis of the CO_2 emissions associated with particular foodstuffs must specify the production-consumption chain involved and establish suitable system boundaries. Thus, for example, an analysis of the carbon content of a kilogramme of strawberries would need data on the carbon intensity of a wide variety of inputs to primary production, including the cultivation of plants, the use of fuel in operating farm machinery, the application of fertilisers and the energy needed to move water for irrigation. It would also need to take into consideration how the soil is prepared, at what time of the year fertiliser is applied and how plant residues are treated. Further issues for consideration would then include the manner in which the strawberries are cleaned, processed, packaged and stored, then transported to the foodservice establishment and prepared. It would also be important to establish the share of strawberries that is finally used by deducting losses due to decay and waste. Indirect energy use would also need to be included; for example in the production of machinery and infrastructure used to produce the strawberries. The calculation is further complicated by various situation-specific factors which may hugely influence the outcome. These include the season (strawberries grown out of season will imply energy use in greenhouses), location (introducing considerations such as local climate conditions, transport distance to the final consumer and mode of transport) and production technique (for example conventional versus organic production). The implications of the last of these considerations can be fundamental. In the case of strawberries, for example, conventional production methods require more fertiliser than organic production per kilogramme of final product. This would lead to additional energy demand for fertiliser production and transport, as well as greater N_2O emissions. Clearly, the calculation of the carbon footprint of even relatively simple products such as strawberries is extremely complicated. The difficulties are magnified for mixed products, such as ketchup or ice cream, making for an even more expensive and time-consuming analysis.

It should be clear from the above example that there are a large number of ways in which foodservice provision can contribute to global GHG emissions. The flipside of this argument is that foodservice establishments actually have considerable room for manoeuvre in engaging with food management practices that will help to mitigate climate change. This section presents a discussion of food management strategies, based on what might be termed the '3 Ps' of climatically sustainable food management: purchasing, preparation and presentation. In each of these categories, foodservice providers have considerable scope to reduce the carbon intensity of the food they provide.

3.1. Purchases

The purchase decision has three main elements from a carbon footprint perspective: the choice of raw materials, their transport and the question of seasonality.

3.1.1. Choosing raw materials

The choice of raw materials can have a considerable impact on emissions. Different food ingredients have different GHG intensities. These differences have been assessed in a review of 54 articles and reports analysing the energy use and related GHG emissions associated with the production and consumption of different foodstuffs, mostly in a European context, by Hille, Ekström, Aall, and Brendehaug (2009). It needs to be noted, however, that comparisons are not always straightforward because the methodologies used are often not consistent, and in some cases, the methodologies are not even specified (Hille et al., 2009; see also Garnett, 2006). There is also considerable scientific uncertainty regarding the relative importance of different GHGs in contributing to climate change, as well as the soil-atmosphere flows of carbon for different forms of agriculture and livestock breeding (Smith et al., 2009). Further problems arise with regard to the assessment of the role of transport, with emissions from aviation not being comparable with those from other transport modes on a CO_2-equivalent basis (Lee et al., 2009).

In view of the difficulties noted above, any comparison of the carbon intensity of different food products should be treated with extreme caution. However, greater confidence can be achieved by comparing food products with the same system boundaries. This allows for some general conclusions to be drawn regarding the GHG intensity of basic foodstuffs. Table 1, for example, presents a review of comparable (production to farm gate) studies of the GHG intensity of vegetables.

The table compares potatoes, carrots, onions, lettuce, tomatoes and cucumbers on a farming lifecycle basis. Several conclusions can be drawn from the comparison. First, even though only a few vegetables are considered, there are considerable differences in emissions. For example, the production of cucumbers in heated greenhouses leads to emissions up to 180 times higher per calorie than the production of tomatoes, onions or potatoes in the field. This suggests that vegetables produced in heated greenhouses should generally be avoided by tourism foodservice providers, although the implications of transport distance and mode still have to be considered.

Cereals, such as wheat, rye, oats and barley, represent the carbohydrate component of many meals. These tend to entail comparably low GHG emissions in their production. The evidence shown in Table 2 suggests that GHG emissions from cereals can vary between 0.180 and 0.720 kg CO_2-e/kg of cereal, or 0.045 and 0.225 kg CO_2-e/1000 kcal (Cederberg, Wivstad, Bergkvist, Mattsson, & Ivarsson, 2005; DEFRA, 2007; Hirschfeld, Weiß, Preidl, & Korbun, 2008; LCA Food, 2003; Stadig, Wallén, & Nilsson, 2001; Tidåker, 2003). Notably, the lowest emissions were found for organic cereal production (Hirschfeld et al., 2008). Rice, on the other hand, would appear to be five to 20 times more emissions-intense, with one study suggesting emissions of 4.55 kg CO_2-e/kg rice, or 1.25 kg CO_2-e/1000 kcal (Kok, Benders, & Moll, 2001). Consequently, it would seem that the use of potatoes and cereals is climatically far more sustainable than using rice, at least in those countries that are producing potatoes and cereals, and where no significant transport emissions share needs to be added on top of this.

One of the most important food categories in terms of GHG content is meat, which generally leads to comparably high emissions as a higher-order food, compared with vegetables. Table 3 reveals that the figures for meat production are less variable than for vegetable production. Thus beef production per 1000 kcal will lead to emissions of about 10 kg CO_2-e, lamb about 8 kg CO_2-e, chicken and turkey 7 kg CO_2-e, and pork 2 kg CO_2-e. Note, however, that various assumptions have been made to allow for comparison, and these are less consistent in the case of chicken (see Lagerberg Fogelberg, 2008; for further explanation of the differences see Hille et al., 2009). Overall, the studies seem to imply that pork is climatically more favourable than chicken, which again is more favourable than lamb and beef. Beef is found to be around five times more GHG-emissions-intense than pork. It should further be noted

Table 1
Greenhouse gas intensity for various vegetables, production on farm.

Vegetable	kg CO_2-e/kg	kg CO_2-e/1000 kcal	Country	Source
Potatoes	0.158	0.247	UK	DEFRA, 2007
	0.160	0.250	Denmark	LCA Food, 2003
	0.261–0.274	0.442–0.464	Netherlands	Kok et al., 2001[a,b]
	0.073–0.083	0.114–0.130	Sweden	Cederberg et al., 2005[a]
	0.100	0.156	Sweden	Mattsson et al., 2001
Carrots	0.046	0.144	UK	DEFRA, 2007
	0.122–0.234	0.381–0.730	Denmark	Miljøstyrelsen, 2006[a,c]
	0.036	0.112	Sweden	Cederberg et al., 2005
Onions	0.060	0.201	Sweden	Cederberg et al., 2005
	0.079	0.265	UK	DEFRA, 2007
	0.382	1.28	Denmark	Miljøstyrelsen, 2006
Lettuce	0.602	5.46	UK	DEFRA, 2007
Tomatoes (greenhouse)	0.082 (unheated)	0.456 (unheated)	Spain	Antón, Montero, & Muñoz, 2005[d]
	1.30	7.20	Sweden	Möller Nielsen, 2007
	5.90–28.50	33.00–158.00	UK	Williams, Audsley, & Sandars, 2006[e]
	3.45–4.92	19.10–27.30	Denmark	Miljøstyrelsen, 2006[a]
Cucumber (greenhouse)	4.37	45.00	Denmark	Miljøstyrelsen, 2006

System boundary: farm production including all greenhouse gases based on lifecycle analysis.
[a] Lower value: conventional, higher value: organic production.
[b] Own calculation to include other GHGs.
[c] Organic production.
[d] Greenhouse, unheated.
[e] Higher values relate to cocktail tomatoes.

that uncertainties persist regarding the role of pastures as carbon sinks (Smith et al., 2009). It is also unclear whether organic or conventional production is more favourable in terms of emissions, with different studies reaching different conclusions regarding this question (Basset-Mens & van der Werf 2005; Hirschfeld et al., 2008).

Another important group of foodstuffs is seafood. Seafood generally has a small carbon footprint, although this depends very much on how the fish is caught, as lifecycle analysis suggests that 75–90% of energy use is related to harvesting (Tyedmers, 2001). For example, per tonne of deep-sea fish, including important commercial species such as cod, fuel use for ships can vary between 230 and 2724 L. For pelagic species, including herring and mackerel, values are lower at 19–159 L/t. The highest fuel use is entailed in shrimp fisheries, with per-tonne fuel use varying between 331 and 2342 L (Tyedmers, 2001). These values affect emissions per calorie of food, which vary between 0.085 kg CO_2-e/1000 kcal for mackerel (Tyedmers, 2001) and 109 kg CO_2-e/1000 kcal for lobster (LCA Food, 2003). The variation is by a factor of 1280, which is by more than three orders of magnitude.

Results from these studies indicate that pelagic fish species such as herring or mackerel entail comparably low GHG emissions per calorie (see Table 4), and that their use in preference to other fish species would consequently make a major contribution to reducing carbon emissions when serving fish. On the other hand, deep-sea species, such as cod, as well as farmed carnivorous fish, such as salmon, should either be avoided or used in lower quantities by tourism foodservice providers. The results also indicate that using low-carbon fish and meats can reduce a meal's average carbon foodprint substantially. It should be remembered, however, that these values are for production only, and that transport may heavily influence the carbon balance (for a discussion of the comparability of these values see Hille et al., 2009).

Table 2
Greenhouse gas intensity for various cereals, production on farm.

Cereal	kg CO_2-e/kg	kg CO_2-e/1000 kcal	Country	Source
Wheat	0.357	0.113	UK	DEFRA, 2007
	0.280–0.710	0.089–0.225	Denmark	LCA Food, 2003
	0.180–0.403	0.057–0.128	Germany	Hirschfeld et al., 2008[a]
Rye	0.620–0.720	0.190–0.225	Denmark	LCA Food, 2003[a]
Rice	4.549	1.250	Netherlands (import)	Kok et al., 2001

[a] Lower value: organic, higher value: conventional production.

3.1.2. Transport

> Everything I fly. Even the coffee I fly. In terms of fuel, this hotel is terrible. We place the order for Christmas stuff now: it will be coming from Dubai, by container. If I order now, I don't have to fly it here. Christmas for me is August latest. You have to plan in a lot of advance if you don't fly the things here. (General Manager, five-star hotel, Seychelles, personal communication, unpublished interview 11 July 2007).

Food imports are, as discussed in the introduction, a particular problem in tropical island states, where foodservice providers are typically serving high-quality, high-protein foods to upscale tourists. Such tourists often, at least in the perception of hotel managers, expect the foodstuffs they know from home (Pattullo, 2005). In such locations, a large share of the food is often imported by air, including food items such as soft drinks, dairy products and even vegetables (Gössling & Schumacher, 2010). While this represents an extreme situation, even in regions such as Europe, the USA or Australia, the transportation of foodstuffs can imply considerable GHG emissions.

There are a number of factors influencing the importance of transport in generating GHG emissions, such as the transport mode chosen. For example, information provided by LRF (2002) indicates that farm production of a kilogramme of potatoes causes emissions of 0.101 kg CO_2-e, the cleaning, packaging and distribution to retailers causes 0.092 kg CO_2-e, and transport to the customer's home and cooking causes 0.145 kg CO_2-e (calculated by Hille et al., 2009). Production thus accounts for only 30% of total GHG emissions, with transport to the consumer's home and cooking accounting for the greatest share. Rather higher ratios can

Table 3
Greenhouse gas intensity of various meats, production on farm.

Meat	kg CO_2-e/kg	kg CO_2-e/1000 kcal	Country	Source
Beef	29.00	16.10	Denmark	LCA Food, 2003[b]
	15.00	8.30	Sweden	LRF, 2002
	19.00–22.00	10.60–12.20	Sweden	Cederberg & Darelius, 2000[c]
	24.00	13.30	Sweden	Cederberg & Nilsson, 2004
	18.00–28.00	10.00–15.60	Ireland	Casey & Holden, 2006
	11.60–18.40	6.40–10.20	Germany	Hirschfeld et al., 2008
	22.30–22.90	12.40–12.70	Germany	Hirschfeld et al., 2008[d]
	18.00	10.00	Netherlands	Kok et al., 2001[b]
	13.40	7.40	Sweden	LRF, 2002
Lamb	ca. 19.00	8.30	UK	DEFRA, 2007[b]; Williams et al., 2006
Pork	3.75	1.50	Denmark	LCA Food, 2003[b]
	4.80	1.90	Sweden	Cederberg & Darelius, 2001
	5.10–8.80	2.00–3.50	France	Basset-Mens & van der Werf, 2005[a,b]
	2.70–4.00	1.10–2.60	Germany	Hirschfeld et al., 2008[a]
	4.25	1.70	Sweden	LRF, 2002
Chicken	ca. 3.70	6.90	Denmark	LCA Food, 2003
	8.17	7.60	Netherlands	Kok et al., 2001[b]
	1.35	1.25	Sweden	LRF, 2002
Chicken and turkey	ca. 7.10–10.30	6.60–9.50	UK	Williams et al., 2006[a]

Conversion factors used: 1 kg beef = 1.800 kcal, 1 kg lamb = 2.300 kcal, 1 kg pork = 2.500 kcal, and 1 kg chicken = 1.080 kcal.
[a] Lower value refers to conventional, higher value to organic production.
[b] Own calculation, involving other sources.
[c] Organic production.
[d] Higher values relate to conventional production, lower values to organic production.

be found for meats, where production accounts for 96% of total GHG emissions in the case of beef, 89% in the case of pork and 75% in the case of chicken (calculated by Hille et al., 2009). Nevertheless, it is clear that transport decisions can considerably affect the GHG intensities of some foods, particularly those with a relatively high weight-to-calorie or volume-to-calorie ratio. Such foods can be costly in energy terms to transport relative to the energy they contain when eaten. Many vegetables, for example, have a low calorific value in relation to their weight and volume. Even so, the importance of this factor depends on transport distances and modes. The shelf life of the foodstuffs being transported may also be a significant consideration, particularly in terms of possible wastage during transit.

The difficulties in calculating the contribution made by transport can be illustrated in an example of local purchases versus longer-distance imports (based on Holtskog, 2001, for Norway). Trucks with a capacity of more than 11 t using on average 62% of their capacity will cause emissions of 0.013 kg CO_2/tkm, while smaller delivery vans with an average capacity usage of 35% will cause emissions of 0.079 kg CO_2/tkm. In a case where a local farmer delivers 150 kg of apples, potatoes or carrots to a nearby foodservice provider, the transport is likely, calculated per tkm, to cause greater emissions over this 25 km than those caused by a heavy truck bringing the same products from 1000 km away. This is due to the greater capacity of the truck. Ultimately, however, the outcome will depend on distribution patterns. For example, if the heavy truck fetches its load from a central storage and delivers it to a wholesaler, this will then demand further transport by smaller transport units to the foodservice establishments.

Table 5 illustrates the energy use entailed in transporting one kilogramme of various foodstuffs by various means of transport (see Appendix A for information on how the calculations were made). The table shows that the transport of melons to Germany from Brazil by boat implies the least kg CO_2/kg of food, due to the greater capacity of the boat in comparison to land or air transport modes, even allowing for the potential for wastage implied by the longer travel time. Grapes by aircraft from South Africa imply the greatest kg CO_2/kg, while transport by air generally is the most CO_2-intensive. The table also allows for a direct comparison between two modes of transporting tomatoes to Germany from Spain, suggesting that their transport by road may be nine times more CO_2-intensive than their transport by rail. This effect is entirely due to the different fuel efficiencies of these two modes of transporting tomatoes over such a long distance.

Table 4
Greenhouse gas intensity of seafood.

Fish species	kg CO_2-e/kg	kg CO_2-e/1000 kcal	Country	Source
"Pelagic fish"	0.060–0.502	0.034–0.286	Various	Tyedmers, 2001
Herring	0.580	0.330	Denmark	LCA Food, 2003
Herring	0.994	0.565	Netherlands	Kok et al., 2001[a]
Mackerel	0.170	0.085	Denmark	LCA Food, 2003
"Deep-sea fish"	0.727–8.61	2.15–25.51	Various	Tyedmers, 2001
	0.784–2.02	2.32–5.99		
Cod	1.20	3.60	Denmark	LCA Food, 2003
Flatfish	3.30	8.60	Denmark	LCA Food, 2003
Shrimps	1.05–7.40	3.17–22.43	Various	Tyedmers, 2001
	2.94	8.90	Denmark	LCA Food, 2003
Lobster	20.20	109.00	Denmark	LCA Food, 2003

[a] Calculation by Hille et al. (2009).

3.1.3. Seasonality

Foodstuffs are seasonal in many regions of the world, especially fruit or vegetables, but also many fish species and some meats, such as lamb or wild game (Hille et al., 2009). Foodstuffs consumed outside their season will need to have been imported or stored. Both imports (see Table 5) and storage can be particularly energy intense. Many months of cold storage will also almost inevitably result in the loss of a quantity of the food, which further increases energy usage and GHG emissions per kcal (Hille, 1998). Attempts by foodservice providers to reduce GHG emissions could therefore focus on ensuring that seasonal foods are not used when they are out of season, except perhaps when there are opportunities to store them at minimal energy cost.

Table 5
Emissions involved in transporting various foodstuffs to Germany.

Food transported (1 kg)	Distance			CO_2	
	Great circle (km)	Corrected (km)[a]	Distance source	kg CO_2/tkm[b]	kg CO_2/kg food
1. Grapes by aircraft from South Africa	8690	9125	http://www.webflyer.com/travel/mileage_calculator/	0.725	6.62
2. Nile perch by aircraft from Tanzania	6980	7329	http://www.webflyer.com/travel/mileage_calculator/	0.725	5.31
3. Strawberries by aircraft from Egypt	2910	3056	http://www.webflyer.com/travel/mileage_calculator/	0.725	2.22
4. Melons by ship from Brazil		9723	http://www.portworld.com/map/	0.015	0.15
5. Tomatoes by truck from Spain		2333	http://maps.google.com/	0.160	0.37
6. Tomatoes by train from Spain		2441	http://maps.google.com/ (shortest rail route)	0.015	0.04

[a] Aircraft cannot usually fly directly between two points, so a detour factor has thus been used to correct for this.
[b] Emission factors based on Gilbert and Perl (2008), Léonardi and Baumgartner (2004), and Zachariadis and Kouvaritakis (2003).

3.2. Preparation

The principal considerations involved in the 'preparation' dimension of food management are firstly how the meal components are chosen and put together, secondly the use of carbon-intensive foodstuffs, especially meats, and thirdly how waste is managed. These three considerations are intimately linked and it is important to recognise that it is not feasible to address any one of them in isolation from the others.

A good starting point for foodservice providers is to reconsider the various components of the meals they serve. The previous subsection has already noted some basic principles, such as reducing the use of anything that has been transported by air or that is for some other reason environmentally or socially problematic (Fearnside, 1997; Primavera, 1997; Rönnbäck, 1999). Companies such as Scandinavian Service Partners Sweden (2009) have already adopted such choice-editing policies. Thus, the company purports to not purchase any fish species that has been identified by the Worldwide Fund for Nature (WWF, 2009) as being threatened. The findings presented in the previous sub-section of this article suggest that as well as helping to address particular environmental and socio-cultural problems, such policies could make a significant contribution to reducing GHG emissions. Similarly, avoiding the use of certain kitchen products, for example aluminium foil, could not only help to address concerns about the environmental impacts of such materials but also make a significant contribution to climatically sustainable operations (Kuckshinrichs & Poganietz, 2006).

A second strategic issue concerns the use of climate-intense foodstuffs, including meats. Seeing meats as a side component rather than the core element of the meal can be a starting point for reducing emissions, and is successfully being used by foodservice providers such as the Salt & Brygga restaurant in Malmö, Sweden (Björn Stenbeck, restaurant owner, personal communication, unpublished interview 24 October 2007). The general rule is that fresh components are usually preferable to semi-produced or processed products. However, care must be taken when cleaning and preparing fresh foodstuffs to ensure that as little as possible is wasted. Energy use for preparing the food is also considerable, but there is evidence that catering kitchens can be more efficient than household kitchens in preparing food, if calculated on a per-meal basis (Carlsson-Kanyama, et al., 2003). Another contribution to reducing emissions could potentially be made if certain products, such as bread rolls, are bought from a factory rather than being produced in the foodservice provider's kitchen (Hille et al., 2009; note that the analysis includes transport from the factory). If such foods are to be produced in-house, the best advice for foodservice providers would be for them to purchase their energy requirements from renewable energy sources, while at the same time seeking to reduce overall energy use in food preparation (for example with regard to cold storage, freezer units, and so on). There are also several useful techniques for analysing energy consumption which foodservice providers could adopt. For restaurant chains, for example, comparing energy use per unit of turnover across restaurants could help them to identify particularly energy-intense cooking routines. With regard to packaging, foodservice providers could also consider the implications of various materials in generating GHG emissions and waste (Kuo, Hsiao, & Lan, 2005).

Finally, waste could be avoided by more carefully planning purchases. Such efforts could be accompanied by collecting and analysing statistics on the amounts of different foods served to different types of guests. For instance, if it were known that the guests for the coming evening are to include a group of pensioners, rather than a teenage sports team (such information could be collected when group bookings are made), the kitchen could be instructed to vary the quantity of each of the various dishes to be prepared according to their anticipated preferences. Another point to consider is whether meals should only be prepared after orders have been placed. Purchases of high-quality foodstuffs can help to increase their shelf life; as can ensuring that they are used to the greatest extent possible, with the minimum of wastage. Storage solutions should also be identified and made available for food that is unused. Finally, it is important to inform all staff about the consequences of waste and to collect food waste separately – for instance by setting up containers for different food fractions – in order to gain a better understanding of which foodstuffs are discarded in which quantities (Swedish Environmental Protection Agency, 2008). This indicates the importance of the training of chefs in hotel schools, where food management for climate mitigation could become a key subject in curricula.

3.3. Presentation

Finally, presentation refers to the way in which foodservice providers serve the meals they have prepared to their customers. The presentation of meals has significant implications for the choice of food components and the manner of their preparation, and these can play a major role in efforts to reduce their carbon footprint. Some foodservice providers may feel that they cannot afford to omit certain foods from their menus, such as seafood or imported beef, considering it preferable to leave the decision of whether or not to purchase a particular dish to the customer. The hope is that customers will naturally steer away from those choices they know to be harmful in contributing to climate change. This strategy does, of course, rely on the consumer firstly having been made aware of the climate implications of their meal and secondly being willing to alter their choice accordingly. Few foodservice providers have yet taken the step of making such information readily available to their patrons, for example by printing it on their menus (for an exception, see Swedish hamburger chain Max, 2010). Indeed, it has been noted that 'social marketing' techniques generally "have remained rarely used in tourism … tourism marketing has concentrated on increasing visitor numbers and market share rather than the more difficult, demanding, and controversial task of changing behaviours" (Peeters, Gössling, & Lane, 2009: p. 249). It is also important that the foodservice provider has a strong grasp of what guests demand, as there may be a mismatch between the perception of restaurant owners and the expectations of guests. Torres (2002), for example, has shown that international tourists in Yucatan, Mexico, may be far more amenable to eating local foods than hotel managers seem to believe. Bélisle (1984) reported that the share of imported foodstuffs varied between 5% and 80% in hotels across Jamaica, suggesting that food import choices are to a large degree made by hotel managers rather than by tourists.

There are, however, some examples of foodservice providers that have experimented with social marketing. For example, the Maritim pro Arte hotel in Berlin serves 140,000 breakfast guests and another 300,000 lunch, dinner and banquet guests per year in its three restaurants (Marcell Kästner, Food and Beverage Manager, personal communication, 5 July 2009). One relevant aspect is the use of buffets to present meals to guests. Observational evidence suggests that, compared to table service, buffets may encourage customers to eat greater quantities of food, eat a greater proportion of climatically more relevant foods, such as meats, and choose environmentally harmful foods that they would not consume at home, such as prawns. Buffet diners also tend to leave more left-overs. Buffets could, on the other hand, aim to inspire customers to eat only the amount of food they wish to consume. In order to achieve this, the Maritim pro Arte offers an alternative organic breakfast buffet with 52 food components, signed with the German "BIO" label for organic food (the conventional buffet has about 100 food components). The alternative buffet is implicitly marketed as a healthier and more high-quality choice compared to the conventional buffet. The hotel also seeks to reduce wastage by providing comparably small plates (with a diameter of only 26 cm). This helps to avoid the 'overloading' plates with food, which diners do not actually want and ultimately remains uneaten. Moreover, carbon-intense foods such as meats are displayed on the periphery of the buffet table, while low-carbon foods are placed at the centre. All items of food are provided in small portions, again with the intention of avoiding wastage.

Taken together, such measures can be significant in influencing consumer choices and have a particular relevance in avoiding waste. The Maritim pro Arte restaurant also offers an organic menu, and the seasonal use of foodstuffs is considered in all meals. Thus, for example, asparagus and strawberries are only provided in the German season, even though they could easily be imported at any time of the year. Guests are said to perceive these efforts in a very positive light (Marcell Kästner, Food and Beverage Manager, unpublished interview, 5 July 2009). These examples correspond with a growing literature that reports that consumer preferences are increasingly turning towards socially and environmentally responsible products (Klein & Dawar, 2004; Mitchell, 2001), with perceptions of value also having considerable importance (Gallastegui & Spain, 2002; Gilg, Barr, & Ford, 2005; McCarty & Shrum, 1994). Crucially, perceptions of value can refer to various aspects of food, including perceived quality, taste, health benefits, animal welfare and responsibilities to future generations. Marketing referring to any of these is more likely to yield emotional responses by customers, creating linkages between sustainability issues (Harper & Makatouni, 2002, Ilbery & Kneafsey, 2000).

Further measures are conceivable, such as to present buffets with small overall amounts of each food component, along with more frequent replenishment of any component that has been used up. Airlines apply such strategies, for instance, with regard to beverages. This could help to avoid wastage, as food from buffets often cannot be stored and re-used. It could be combined with an 'always fresh' marketing strategy, which explains that new food is continuously provided for quality reasons. With regard to à la carte menus, it may often be possible to 'convince' customers to choose particular, more climatically sustainable dishes, for instance through recommendations by waiting staff.

4. Discussion and conclusions

The findings presented in this article have confirmed the need to study the role of food in tourism in more depth, as indeed has been demanded by various researchers for the past 25 years (Bélisle 1983; Telfer & Wall 1996, 2000). There are a number of important inter-relationships between food production and consumption in tourism that are of critical relevance to sustainability. These include a vital economic dimension: particularly in view of claims that as much as one third of tourism spending may be on food (Torres, 2000). This article has outlined the relationships between food production, consumption and climate change. In doing so, it has indicated the role that food management could play in making a critical contribution to mitigating climate change.

While more research is needed to obtain a better understanding of the GHG intensity of different foodstuffs, there are some general findings that could inspire foodservice providers of all kinds to engage in more climatically sustainable food management. Table 6 summarises a number of principles for providing climatically more sustainable foods, divided into what might be called the '3 Ps' of food management: purchases, preparation and presentation.

With regard to purchases, recommendations would include 'buy as little as possible policies', including vegetables grown in heated

Table 6
Recommendations for climatically sustainable food management.

Purchases	RED – buy as little as possible policy • Buy as little as possible vegetables grown in heated greenhouses • Buy as little as possible foods involving air transport • Buy as little as possible specific species, such as giant, king and tiger prawns, lobster • Buy as little as possible imported beef • Buy as little as possible aluminium foil AMBER – buy less policy • Buy less beef • Buy less deep-sea fish (e.g. cod) • Buy less farmed carnivorous fish (e.g. salmon) • Buy less rice • Buy less seasonal foods out of their season/storage time- GREEN – buy more policy • Buy more locally produced foods, if transported over short distances using CO_2-efficient modes • Buy more potatoes • Buy more grains (including pasta) • Buy more pelagic fish • Buy more pork • Buy more chicken • Buy more foodstuffs with longer shelf lives
Preparation	• Purchase energy from renewable sources • Use more energy-efficient cooking routines • Do not prepare energy-intensive foods in-house • Put dishes on the menu that use less meat and more vegetables • Prepare meals only after orders have been placed • Plan purchases to avoid waste • Separate food waste from general waste
Presentation	• Always present at least one attractive vegetarian alternative • Reduce portion sizes at buffets, with more regular replenishment • Reduce plate size at buffets • Arrange buffets so that less carbon-intensive foods are at the centre • Train staff to recommend less carbon-intensive dishes • Avoid single-use packaging

greenhouses, foods involving air transport, problematic species such as lobster and imported beef, and environmentally harmful materials such as aluminium foil or Styrofoam packages for taking home leftovers. 'Buy less policies' would seek to generally reduce the amount of beef, deep-sea and farmed carnivorous fish, rice, and seasonal foods out of their season. 'Buy more policies' would seek to use greater amounts of locally produced foods (transported over short distances, using CO_2-efficient modes), potatoes and grains, pelagic fish, pork and chicken, as well as foodstuffs with longer shelf lives in order to reduce wastage.

With regard to preparation, foodservice providers are advised to purchase energy from renewable sources, to look into energy-efficient cooking routines, not to prepare energy-intensive foods in-house (such as bread rolls), to put dishes on the menu that use less meat and more vegetables, to prepare meals only after orders have been placed, to plan purchases to avoid waste, and to separate food waste from general waste.

Presentation might include reduced portion sizes and plate sizes at buffets, the arrangement of buffets so that less carbon-intensive foods are at the centre, the training of staff to recommend less carbon-intensive meals, vegetarian alternatives on the menu, and the avoidance of single-use packaging. Compared to business-as-usual food purchasing, preparation and presentation practices, the authors estimate that these measures could save 50–80% of greenhouse gas emissions, indicating that there is considerable scope for food management to make a significant contribution to climate mitigation.

One aspect that has not so far been considered in depth is whether organic food production can make a significant contribution to climate change mitigation. This is because published empirical studies are not generally consistent with regard to the measurement of GHG emissions from organic food production. In the case of vegetables, for example, there are articles suggesting that conventional production entails lower GHG emissions in the case of potatoes (Cederberg et al., 2005; Kok et al., 2001), while others find that organic production may result in lower emissions in the case of carrots (Cederberg et al., 2005). Others studies, however, indicate no significant differences between organically and conventionally produced vegetables (Mattsson, Wallén, Blom, & Stadig, 2001). Further inconsistencies are apparent in respect of cereals and meats (Basset-Mens & van der Werf, 2005; Hirschfeld et al., 2008).

This is not to argue that organic food has no relevance in food management, which may be considered important because of the other benefits associated with it, such as the avoidance of chemicals, fertilisers and irrigation, regardless of their implications for GHG emissions. In addition, organic food could be valued because of the benefits it brings in terms of animal welfare and employee health. Organic food, usually focusing on regional small-scale production, is also an important force countering trends of globalisation and industrialisation of food production processes, which helps to avoid problems incurred in mass production such as food safety (Vos, 2000), large-scale conversion of ecosystems such as rainforests for soybean or palm oil production (Costa, Yanagi, Souza, Ribeiro, & Rocha, 2007; Morton et al., 2006) and food technology issues (Chan & Lai, 2009). Organic farming systems can also help to maintain landscapes and biodiversity (Norton et al., 2009) and to ensure that a greater share of the profits of food production is captured by the farmer.

Similar uncertainties exist regarding the carbon footprints involved in the production of local foodstuffs, particularly with regard to transport. In the context of developing countries, Torres (2002: p. 283) observes that "developing backward linkages to local agriculture is critical in reducing foreign exchange leakages and increasing local multiplier effects". These and other sustainability benefits for organic and locally produced foods may be identified. For instance, Sims (2009) discusses the role of local food in creating a sense of place and authenticity. Meanwhile writers such as Gyimóthy and Mykletun (2009) identify the important role that local foods, can play in generating tourist experiences.

In putting the above recommendations into context, it is important to consider the role of food culture in determining the feasibility of specific food management practices. Thus, for example, the prescription for foodservice providers to 'buy less rice' may be considered less feasible in many Asian countries, where rice often represents the major component of the national diet and has a much greater cultural significance than in the European context. The opposite might be said in respect of the consumption of meat. Economic considerations are also likely to be important in determining the feasibility of particular food management practices. Foodservice providers are, almost without exception, led by the profit imperative. Some food management practices will not make commercial sense in their own right and as such are unlikely to be widely adopted. Other food management practices are, however, likely to contribute to, rather than detract from, the commercial bottom line. This contribution may be direct, such as when practices are introduced to minimise the wastage of food, or indirect, such as when a foodservice provider's business reputation is enhanced by their commitment to sustainable food management.

In concluding this article, the claim that too little research has been carried out on tourism and food sustainability is confirmed. Most of the research that has been published thus far focuses either on the tropical destinations, or else on the UK, where the topic has received significant interest in the context of regional development. There is a general lack of research covering issues such as the GHG intensity of different foodstuffs, consumer perceptions of climatically more sustainable foods and the willingness of consumers to pay premiums for organic or locally produced food. Such research would need to be complemented with supply-side perspectives – for example, how foodservice providers can present and market more sustainable foods – along with the extent to which sustainable food might entail additional costs. This is also clearly an important educational issue for hotel and restaurant schools. Such research can make an important contribution to climate change mitigation, as well as sustainability more generally.

Appendix A. Emission factor calculations

Road transport:

Léonardi and Baumgartner (2004) for the UK, based on sampling data from 153 trucks: trucks < 40 t: 0.182 kg CO_2/tkm; >40 t: 0.080 kg CO_2/tkm.

Zachariadis and Kouvaritakis (2003) for the Eastern European countries (whole fleet estimates for 2010):

- Trucks: 46.8 toe*/Mtkm
- Train transport: 4.9 toe*/Mtkm

*1 toe is equivalent in energy terms to 0.974 t diesel. Diesel has 85.9% carbon content by weight so the emission factor will be $0.859 \times 3.6667 = 3.15$ kg CO_2/kg diesel.

Consequently, emissions are:

- Trucks: 0.144 kg CO_2/tkm
- Train transport: 0.015 kg CO_2/tkm

Gilbert and Perl (2008):

- Air international: 9.9 MJ/tkm
- Marine international: 0.2 MJ/tkm
- Rail: 0.2 MJ/tkm

- Lorry: 2.6 MJ/tkm

1 kg of diesel/kerosene corresponds to 43 MJ of energy, and 3.15 kg CO_2. This translates into:

- Air international: 0.725 kg CO_2/tkm
- Marine international: 0.015 kg CO_2/tkm
- Rail: 0.015 kg CO_2/tkm
- Lorry: 0.190 kg CO_2/tkm

From the above studies, the following average values have been used:

- Air international: 0.725
- Marine international: 0.015
- Lorry: 0.16

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